

Towards Traffic Modelling of Multi-Agent LLM Systems: The Role of Coordination Topology

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Why study multi-agent LLM traffic?

- *“Internet traffic from AI agents and other bots now surpasses that of people”*
— Cloudflare, June 2026
- **Historical precedent:** Web, P2P, video streaming – each new application class demanded its own traffic model.
Leland et al. 1994 · Paxson & Floyd 1995 · Crovella & Bestavros 1997
- **LLM traffic is different:** one user task triggers an **indeterminate** number of internal LLM calls – agents reason, branch, retry, and coordinate.

This work is the first attempt to introduce traffic modelling to agentic systems.

Research question and objectives

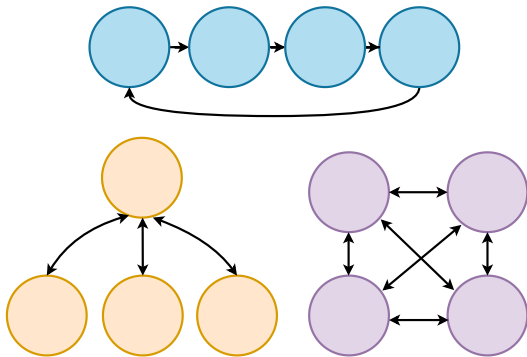
Central question

How does coordination topology shape the traffic model of multi-agent LLM systems?

Framing

- **Independent variable:** coordination topology
 - **Baseline:** do classical Poisson arrival models hold for agentic workloads?
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- ① **Empirical characterisation** of topology-induced arrival processes
 - ② **Structural signatures:** identify fan-out bimodality in arrival distributions
 - ③ **Reject the Poisson null model** and identify better-fitting alternatives
 - ④ **Open-source measurement framework** for reproducible agentic traffic measurement

Multi-agent coordination topologies



Sequential

Agents contribute one at a time

Star

Central orchestrator fans out to parallel reviewers

Full Mesh

Every agent communicates with every other agent

Controlled experiment

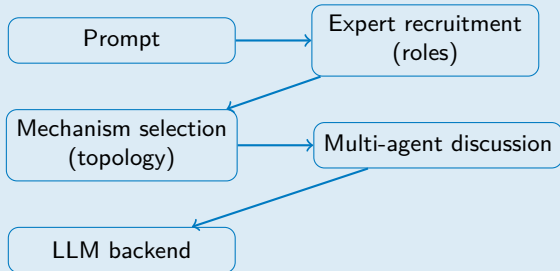
Same task, same model, same backend, same instrumentation – **only topology changes**

Workflow and what we measure

Sample prompt (1 line)

"Research the current state of quantum computing and its potential applications in cryptography."

AgentVerse workflow (per run)



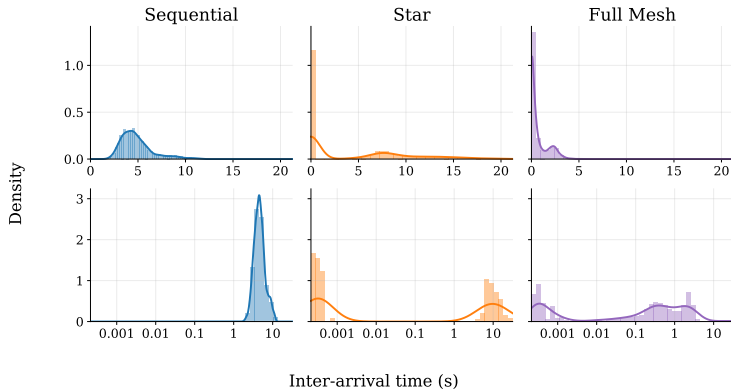
What we ran

- One discussion task repeated **500 times** per topology
- Three topologies: sequential, star, full mesh
- Same model/backend and same instrumentation (empirical testbed)

Metric

IAT: time between consecutive LLM requests observed at the backend (within a run).

Inter-arrival time (IAT) during discussion (by topology)



Sequential

Unimodal, multi-second
gaps

Star

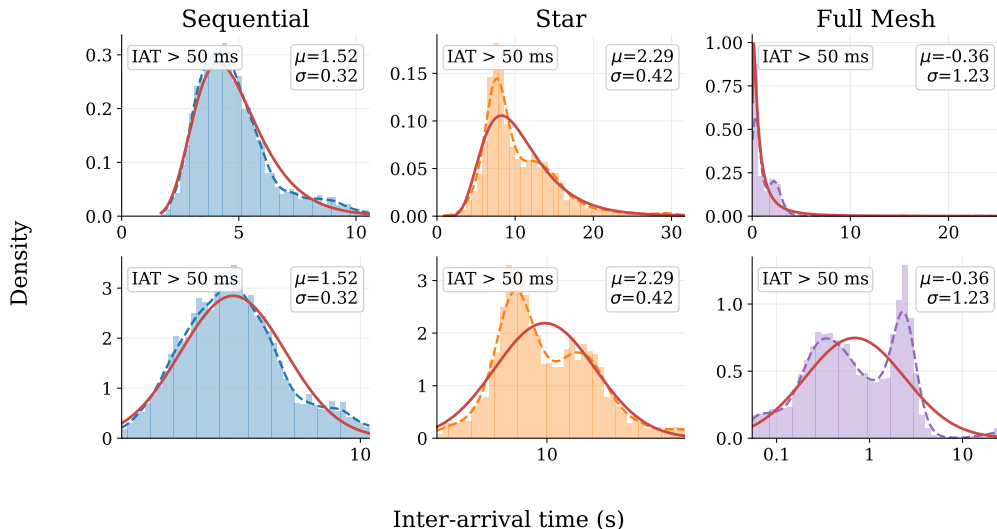
Bimodal: dispatch +
reasoning

Full Mesh

Bimodal, compressed in
time

Mechanism: fan-out coordination creates near-zero IAT bursts + a slower reasoning-phase mode.

Goodness of fit tests (reasoning-only IATs)



Robustness check: subsampling stability

Why subsample?

At large N , the KS test rejects any fixed distribution. We use 500 subsamples of each size n to generate hypothesis tests for each distribution.

n	Distribution	Sequential	Star	Full Mesh
50	Exponential	100%	100%	51%
	Weibull	7%	10%	7%
	Log-normal	1%	2%	1%
100	Exponential	100%	100%	80%
	Weibull	40%	55%	32%
	Log-normal	1%	11%	5%
200	Exponential	100%	100%	98%
	Weibull	93%	98%	76%
	Log-normal	1%	50%	33%

Cells show % of subsamples (size n) where KS rejects H_0 ($\alpha = 0.05$).

H_0 : data follows the specified distribution

- Exponential rejected **consistently** across all topologies and sizes
- Log-normal remains the **most stable** fit

Discussion: implications for infrastructure

Two systems with the **same average task rate** can place **very different load** if they use different topologies

Topology is already studied for:

- Task quality (MARBLE)
- Token cost (G-Designer)

→ **Traffic shape** is a third dimension

Practical implications

- Rate limiters and load balancers need **topology awareness**
- Burst-sensitive metrics (IAT percentiles, burst fraction, concurrency peaks) complement aggregate throughput
- Without knowing the workflow topology, a bursty trace may look like load variation when it is a **structural feature**

Same task, same model: topology alone reshapes the traffic

- IAT distribution varies greatly under different coordination mechanisms; Fan-out makes arrivals bimodal
- Poisson typically rejected, log-normal fits best
- Rate limiters, load balancers, and autoscalers need topology awareness
- Measurement framework is open sourced



Paper



Code

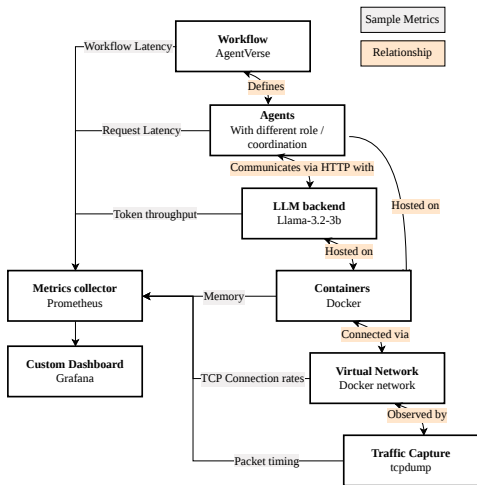
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github.com/dlamagna/agentraffic

Open to discuss PhD positions and funding opportunities

A1: measurement framework architecture



Four observation layers:

- 1 **Application:** call latency, IAT, token counts
- 2 **LLM backend:** concurrency, throughput, queueing
- 3 **Container:** CPU, memory, I/O
- 4 **Network:** TCP flows, byte rates, connection events

Stack: AgentVerse + AsyncVLLM (Llama-3.2-3B) + Prometheus + Jaeger + cAdvisor + Grafana

Modular by design: each layer can be swapped independently. Any agentic framework can sit above this stack and have its traffic characterised using the same instrumentation.

A2: bimodality from fan-out coordination

Structural, deterministic fan-out ($IAT \leq 50$ ms)

Star: orchestrator dispatches parallel reviewer calls

Full mesh: agents send peer messages concurrently

Reasoning-phase component ($IAT > 50$ ms)

Time spent generating tokens, governed by prompt length, context size, and batching

- Sequential has **no fan-out** $\rightarrow$ unimodal
- Star: 54.9% of gaps below 50 ms
- Full mesh: 38.2% below 50 ms
- 50 ms threshold matches the visible trough between modes

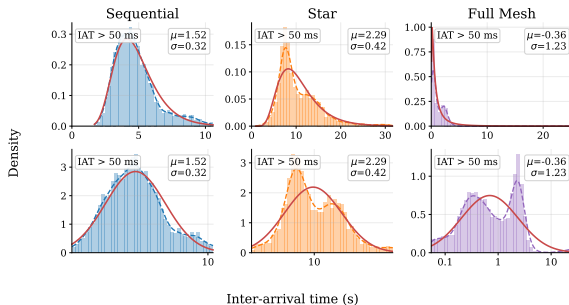
**Structural and deterministic:
topology drives the burst fraction.**

A3: cross-layer comparison

Metric	Sequential	Star	Full Mesh
LLM requests/run	14.03	16.22 (+16%)	29.99 (+114%)
Discussion call rate	0.21 c/s	0.21 c/s (0%)	1.37 c/s (+552%)
Duration	81.1 s	105.0 s (+30%)	31.6 s (-61%)
Burst fraction (IAT <50ms)	0.0%	54.9% (+55 pp)	38.2% (+38 pp)
Median IAT	4.49 s	0.40 ms (-100%)	236 ms (-95%)
IAT p_{95}	8.52 s	16.20 s (+90%)	2.82 s (-67%)
Mean LLM concurrency	0.97	1.62 (+67%)	2.94 (+203%)
Peak LLM concurrency	1.00	4.00 (+300%)	5.00 (+400%)
Prompt-token throughput	237 tok/s	318 tok/s (+34%)	1,052 tok/s (+344%)
Mean LLM TCP byte rate	617 B/s	768 B/s (+25%)	1,090 B/s (+77%)

Topology completely changes the load shape – even for the same task.

A4: Goodness of fit tests



Log-normal fit on reasoning-phase IATs

AIC: observed likelihood under the model, penalised for parameter count

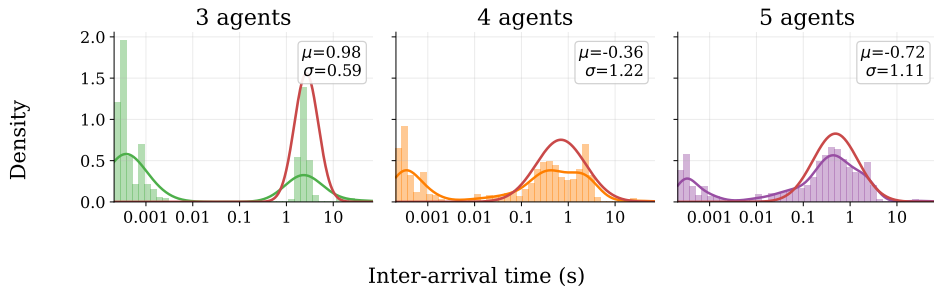
KS: max gap between empirical and fitted CDF

Three candidate families (MLE fit):

	AIC ↓	KS ↓
<i>Sequential</i>		
Exponential	31,604	0.396
Weibull	23,165	0.091
Log-normal	21,782	0.027
<i>Star</i>		
Exponential	22,173	0.359
Weibull	19,131	0.104
Log-normal	18,377	0.077
<i>Full Mesh</i>		
Exponential	22,078	0.116
Weibull	21,760	0.068
Log-normal	20,590	0.089

Log-normal wins across all topologies. Exponential decisively rejected.

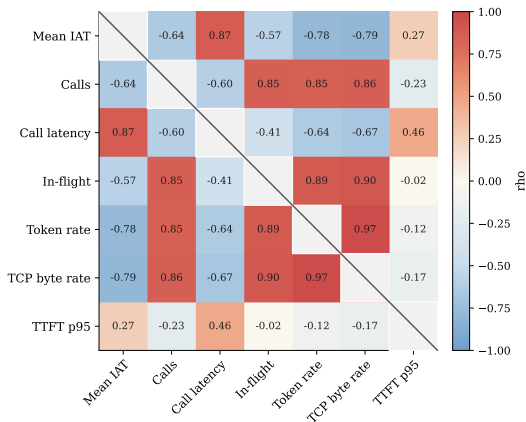
A5: full mesh scaling: varying subagent count



	3	4	5
Calls/run	12.8	31.8	55.3
Tokens/run	28.9k	38.1k	89.7k
Burst frac.	65.5%	37.9%	27.9%
Med. IAT	1 ms	239 ms	307 ms

- Bimodal shape **persists** as agents scale
- Burst fraction **decreases**: reasoning-phase exchanges dominate at higher agent counts
- *Future work*: what happens as $n \rightarrow \infty$?

A6: cross-layer correlation



- Token throughput, in-flight requests, and TCP byte volume **move together**
- Larger IATs negatively correlated with backend traffic rate
- Four measurement layers track the **same topology-induced changes**

Takeaway

Topology effects at the application layer **propagate** to inference concurrency and network-level traffic